EODP Unitary Test Reports

Earth Observation Data Processing
Master in Space Engineering, UC3M

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1 Introduction

1.1 Purpose

This document reports the unitary tests of the EODP simulator. The simulator models a passive optical multi-band imager with four VNIR bands. It has three modules. The Instrument Module (ISM) converts the input scene into digital numbers. The Level-1B module (L1B) converts the digital numbers back into radiances. The Level-1C module (L1C) projects the L1B radiances onto the MGRS grid.

Each test has its own section with the objective, the inputs, the execution, the results, the answer to the theoretical question and the conclusion. The algorithms follow the ATBD [3]. The deliverables follow the guide to the subject [4].

1.2 Tests

Table 1 lists the tests.

Table 1: List of tests.

Test	Module	Section
EODP-TS-ISM	Instrument Module	3
EODP-TS-L1B	Level-1B processing	4
EODP-TS-L1C	Level-1C processing	5
EODP-TS-E2E	End-to-end chain ISM, L1B and L1C	6

1.3 Pass criterion

The outputs of the ISM, L1B and L1C tests are compared with the reference outputs of the teacher. A test passes when the maximum relative difference of every output is lower than 10^{-3} % [4]. The relative difference of each file is

$$\varepsilon = \max_{i,j} \frac{\left| x_{ij} - x_{ij}^{\text{ref}} \right|}{\left| x_{ij}^{\text{ref}} \right|} \cdot 100\,[\%]. \tag{1}$$

Pixels where the reference is zero are compared with the absolute difference. The E2E test has no reference output. It is checked with the radiometric consistency of the chain.

1.4 Test environment

Table 2 shows the environment used to produce the results. The source code is available in GitHub [2].

Table 2: Test environment.

Item	Version
Operating system	`Windows-10-10.0.26200-SP0`
Python	`3.11.9`
NumPy	`2.4.6`
SciPy	`1.17.1`
Matplotlib	`3.11.2`
netCDF4	`1.7.5.1.post1`
mgrs	`1.5.0`
pyproj	`3.7.2`
Git commit	`39626a8`

1.5 References

- [3] EODP Algorithm Theoretical Baseline Document, revision 1.4.
- [4] Guide to the Subject, revision 3.0.

2 Test harness

2.1 Structure

The harness has one folder per test inside `harness/`. Each folder contains the four items required by the guide [4].

- `test_*.py`. The script that launches the test and generates the figures and tables.
- `log/`. The log of the test script and the logs of the EODP modules of the final execution.
- `output/`. All the outputs generated in the execution.
- `figures/`. All the figures of the test. They are the same figures included in this report.

The folder `harness/common/` has the shared functions. They handle the paths, the logs, the figures and the cross-validation. The cross-validation tables of this report are written by the scripts.

```
harness/
  common/            harness_utils.py, environment.py
  EODP-TS-ISM/       test_ism.py, log/, output/, figures/
  EODP-TS-L1B/       test_l1b.py, log/, output/, figures/
  EODP-TS-L1C/       test_l1c.py, log/, output/, figures/
  EODP-TS-E2E/       test_e2e.py, log/, output/, figures/
```

2.2 Execution

The scripts run from the root of the repository with the Python environment of Table 2. The input data of the teacher is placed in `EODP_TER_2021/` and `EODP-TS-E2E/`.

```
python harness/EODP-TS-ISM/test_ism.py
python harness/EODP-TS-L1B/test_l1b.py
python harness/EODP-TS-L1C/test_l1c.py
python harness/EODP-TS-E2E/test_e2e.py
```

The own outputs are written in the harness folders. The reference outputs are never overwritten.

2.3 Changes to the provided code

The skeleton of the simulator was completed with the algorithms of the ATBD [3]. Some changes were needed to run it in the test environment.

- The main scripts use paths relative to the repository.
- The configuration files are loaded with a small loader. If the variable `EODP_CONFIG_DIR` is set, the configuration is read from that folder. The E2E test uses it to load the E2E configuration.
- The unused `interp2d` imports were removed. The function no longer exists in recent SciPy versions.
- The `mgrs` library is fixed to version 1.5.0. Later versions return the centre of the MGRS tile instead of its south-west corner. The reference outputs use the corner.

2.4 Traceability to the ATBD

Table 3 links each algorithm of the ATBD [3] with the function that implements it.

Table 3: Algorithms of the ATBD and their implementation.

ATBD identifier	Algorithm	Function
EODP-ALG-ISM-1010	Spectral modelling, ISRF	`opticalPhase.spectralIntegration`
EODP-ALG-ISM-1020	Radiances to irradiances	`opticalPhase.rad2Irrad`
EODP-ALG-ISM-1030	Spatial modelling, system MTF	`mtf.system_mtf` `opticalPhase.applySysMtf`
EODP-ALG-ISM-1031	Diffraction MTF	`mtf.mtfDiffract`
EODP-ALG-ISM-1032	Defocus MTF	`mtf.mtfDefocus`
EODP-ALG-ISM-1033	Wavefront aberration MTF	`mtf.mtfWfeAberrations`
EODP-ALG-ISM-1034	Detector MTF	`mtf.mtfDetector`
EODP-ALG-ISM-1035	Smearing MTF	`mtf.mtfSmearing`
EODP-ALG-ISM-1036	Motion blur MTF	`mtf.mtfMotion`
EODP-ALG-ISM-2010	Irradiances to photons	`detectionPhase.irrad2Phot`
EODP-ALG-ISM-2030	Photons to electrons	`detectionPhase.phot2Electr`
EODP-ALG-ISM-2020	PRNU	`detectionPhase.prnu`
EODP-ALG-ISM-2040	Dark signal	`detectionPhase.darkSignal`
EODP-ALG-ISM-2050	Bad and dead pixels	`detectionPhase.badDeadPixels`
EODP-ALG-ISM-3010	Electrons to voltage	`videoChainPhase.electr2Volt`
EODP-ALG-ISM-3020	Digitisation	`videoChainPhase.digitisation`
EODP-ALG-L1B-1010	Equalization	`l1b.equalization`
EODP-ALG-L1B-1020	Restoration	`l1b.restoration`
EODP-ALG-L1C-1000	Orthorectification onto MGRS	`l1c.l1cProjtoa`

3 EODP-TS-ISM. Instrument Module

3.1 Objective

The test validates the Instrument Module. The ISM converts the input TOA spectral radiances into digital numbers in three stages [3].

- Optical stage. Spectral integration with the ISRF, conversion to irradiance and application of the system MTF.
- Detection stage. Conversion to photons and electrons, PRNU, dark signal and bad or dead pixels.

- Video chain. Conversion to volts and digitisation.

The test also answers two questions. Which MTF defines the system, and do the unit conversion factors reproduce the saved outputs.

3.2 Inputs and configuration

The input is the scene `gradient_alt100_act150/sgm_toa.nc`. It has 100 lines ALT, 150 columns ACT and 600 wavelengths from 400 to 999 nm. The radiance grows linearly in the ACT direction. Table 4 lists the main parameters of the default configuration.

Table 4: Main ISM configuration parameters.

Parameter	Symbol	Value
Pixel size	w	30 µm
Integration time	t_{int}	6.72 ms
Pupil diameter	D	0.150 m
Focal length	f	0.5262 m
Optical transmittance	T_r	0.99
Central wavelengths	λ	490, 665, 865, 945 nm
Quantum efficiency	QE	0.8 e^-/ph
Full well capacity	FWC	420000 e^-
Output conversion factor	OCF	5.4×10^{-6} V/e^-
ADC gain	G	0.56
Bit depth	N_{bits}	12
Voltage range	$V_{max} - V_{min}$	0.86 V
PRNU and DSNU factors	k_{PRNU}, k_{DSNU}	0.04, 0.2
Bad and dead pixels		1.0 %, 0.5 %

3.3 Execution

The script `harness/EODP-TS-ISM/test_ism.py` runs the ISM with all the effects enabled. It saves every intermediate output and the seven MTFs of each band. Then it compares them with the reference outputs and generates the figures and tables of this section.

3.4 Results

3.4.1 Cross-validation

Tables 5 and 6 show the maximum relative difference of every output with respect to the reference. All the files are identical to the reference. This includes the PRNU and the dark signal, which use random numbers with a fixed seed.

Table 5: ISM cross-validation of the TOA outputs. Maximum relative difference [%] per band.

File	VNIR-0	VNIR-1	VNIR-2	VNIR-3	Result
`ism_toa_isrf_`	0	0	0	0	Pass
`ism_toa_optical_`	0	0	0	0	Pass
`ism_toa_e_`	0	0	0	0	Pass
`ism_toa_prnu_`	0	0	0	0	Pass
`ism_toa_ds_`	0	0	0	0	Pass
`ism_toa_detection_`	0	0	0	0	Pass

File	VNIR-0	VNIR-1	VNIR-2	VNIR-3	Result
`ism_toa_`	0	0	0	0	Pass

Table 6: ISM cross-validation of the MTFs. Maximum relative difference [%] per band.

File	VNIR-0	VNIR-1	VNIR-2	VNIR-3	Result
`Hdiff_`	0	0	0	0	Pass
`Hdefoc_`	0	0	0	0	Pass
`Hwfe_`	0	0	0	0	Pass
`Hdet_`	0	0	0	0	Pass
`Hsmear_`	0	0	0	0	Pass
`Hmotion_`	0	0	0	0	Pass
`Hsys_`	0	0	0	0	Pass

3.4.2 Signal through the stages

Figure 1 shows the central ALT line after each stage. The radiance after the ISRF is a straight line, as the input scene. The MTF modifies the first and last columns. The FFT treats the image as periodic, so the bright last column is mixed with the dark first one. The detection stage adds the stripes of the PRNU and the DSNU. In VNIR-2 and VNIR-3 the signal reaches the full well capacity of 420000 e^-. After the video chain VNIR-1, VNIR-2 and VNIR-3 saturate at 4095 DN.

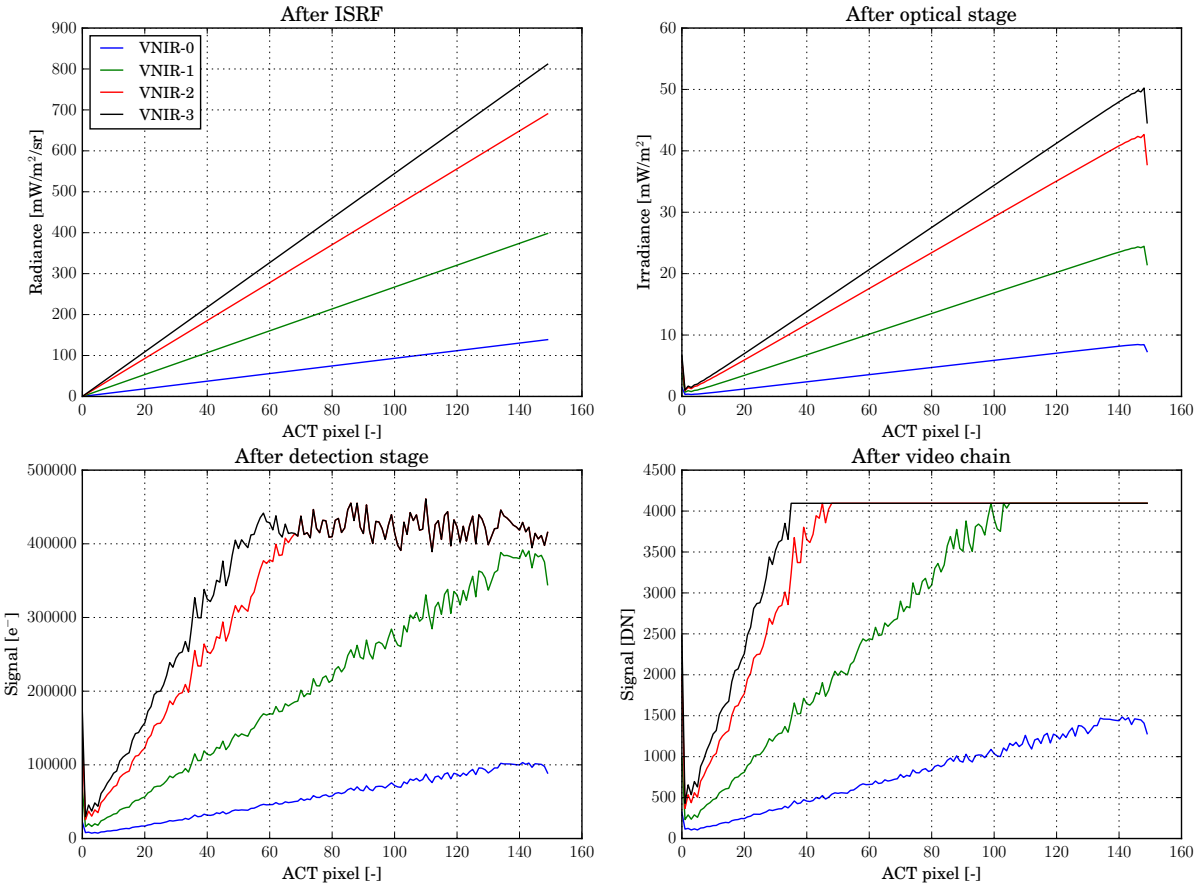


Figure 1: ISM. Central ALT line after each stage for the four bands.

3.4.3 MTF

Figures 2 to 5 show the cuts of the MTF in the ACT and ALT directions up to the Nyquist frequency. The frequencies are normalised with the inverse of the pixel width, so Nyquist is at 0.5. Table 7 gives the value of each contributor at the last sample before Nyquist.

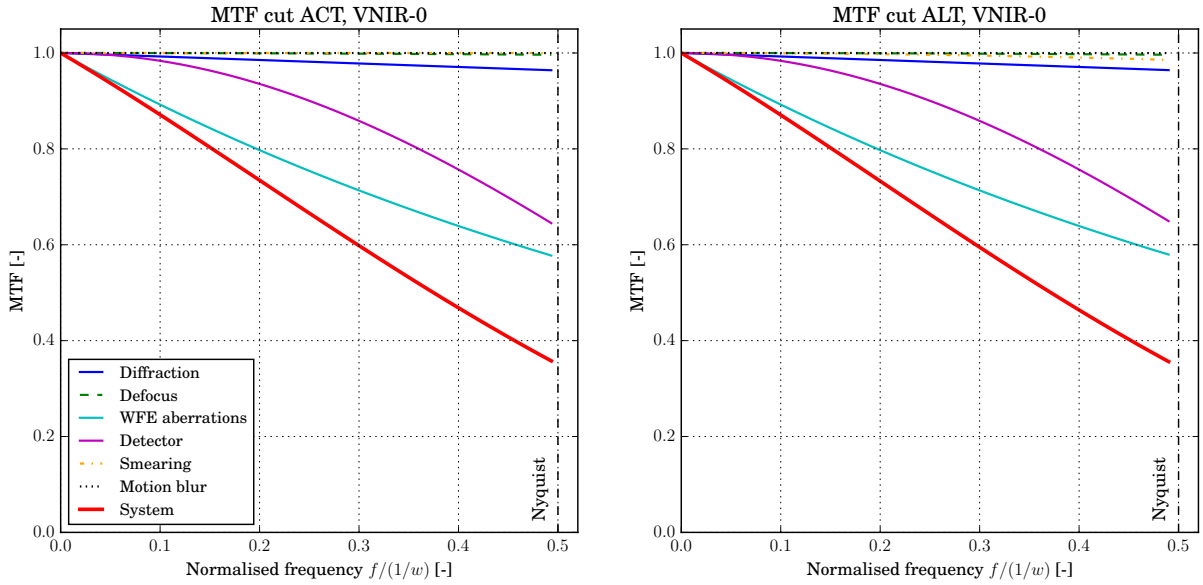


Figure 2: ISM. MTF cuts ACT and ALT, VNIR-0.

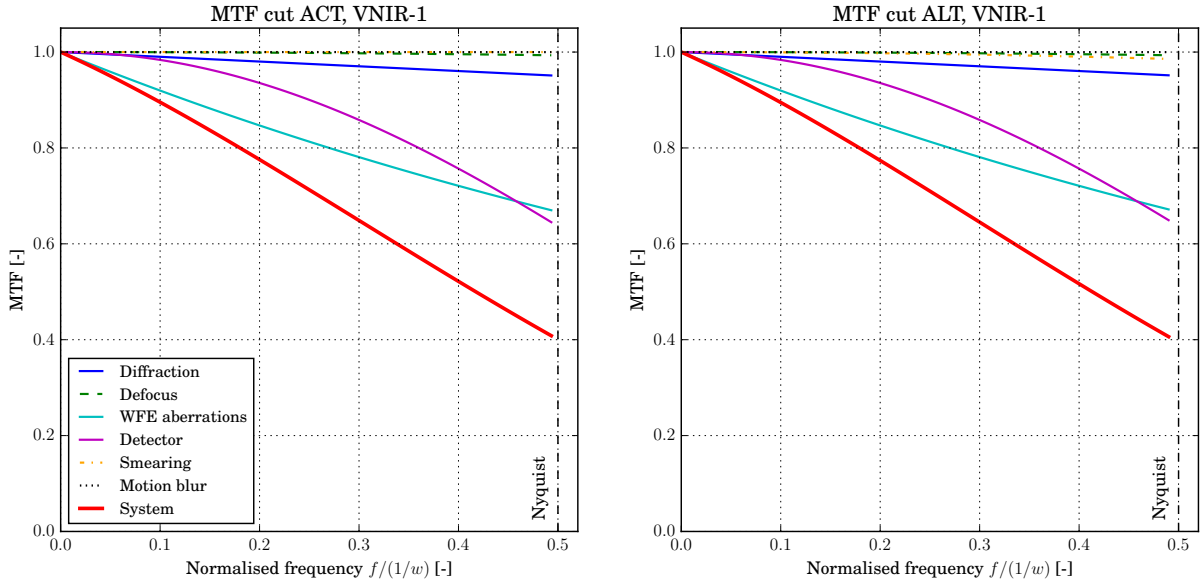


Figure 3: ISM. MTF cuts ACT and ALT, VNIR-1.

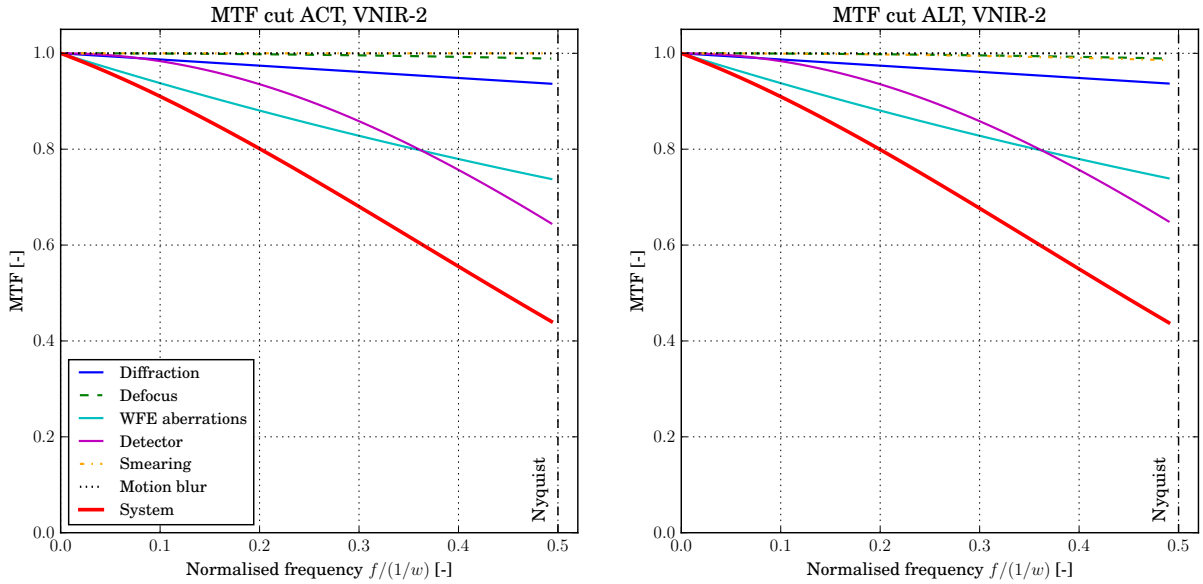


Figure 4: ISM. MTF cuts ACT and ALT, VNIR-2.

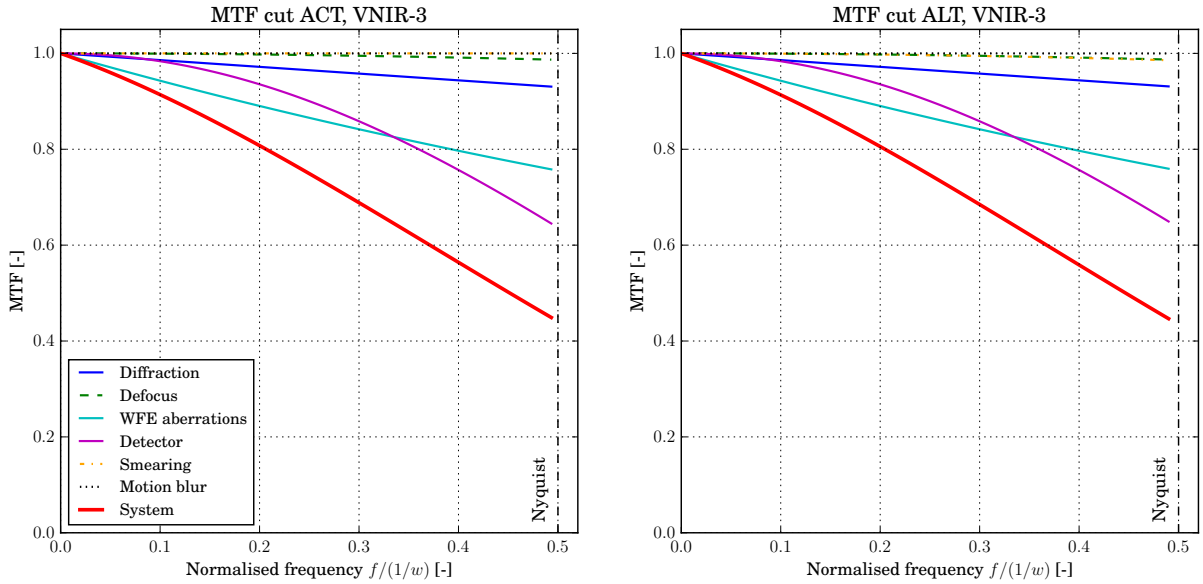


Figure 5: ISM. MTF cuts ACT and ALT, VNIR-3.

Table 7: MTF of each contributor at the last sample before Nyquist.

	VNIR-0		VNIR-1		VNIR-2		VNIR-3	
MTF	ACT	ALT	ACT	ALT	ACT	ALT	ACT	ALT
Diffraction	0.964	0.964	0.951	0.951	0.936	0.937	0.931	0.931
Defocus	0.996	0.996	0.993	0.993	0.989	0.989	0.987	0.987
WFE aberrations	0.577	0.580	0.670	0.672	0.738	0.739	0.758	0.759
Detector	0.645	0.649	0.645	0.649	0.645	0.649	0.645	0.649
Smearing	1.000	0.986	1.000	0.986	1.000	0.986	1.000	0.986
Motion blur	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
System	0.358	0.356	0.408	0.406	0.441	0.438	0.449	0.447

3.4.4 Unit conversion at the central pixel

Each conversion of the ISM is a scalar factor for a given band. Table 8 lists the factors. They are

$$I = L \cdot T_r \frac{\pi}{4} \left(\frac{D}{f}\right)^2, \qquad N_{ph} = I \cdot \frac{A_{pix}\, t_{int}\, \lambda}{hc}, \qquad N_e = N_{ph} \cdot QE, \tag{2}$$

$$V = N_e \cdot OCF \cdot G, \qquad DN = \text{round}\left(V \cdot \frac{2^{N_{bits}} - 1}{V_{max} - V_{min}}\right). \tag{3}$$

The irradiance is converted from mW to W before the conversion to photons. The factors are applied to the radiance of the central pixel [50, 75] of `ism_toa_isrf`. The result is compared with the two outputs that the code saves in these units. They are the electrons of `ism_toa_e` and the digital numbers of `ism_toa`. Irradiance, photons and volts are not saved. Table 9 shows the comparison.

Table 8: Unit conversion factors per band. The two largest factors of each band are in bold.

Step	Units	VNIR-0	VNIR-1	VNIR-2	VNIR-3
Radiance → irradiance	[sr]	6.32×10^{-2}	6.32×10^{-2}	6.32×10^{-2}	6.32×10^{-2}
Irradiance → photons	[ph m^2/mW]	$\mathbf{1.49 \times 10^4}$	$\mathbf{2.02 \times 10^4}$	$\mathbf{2.63 \times 10^4}$	$\mathbf{2.88 \times 10^4}$
Photons → electrons	[e$^-$/ph]	8.00×10^{-1}	8.00×10^{-1}	8.00×10^{-1}	8.00×10^{-1}
Electrons → volts	[V/e$^-$]	3.02×10^{-6}	3.02×10^{-6}	3.02×10^{-6}	3.02×10^{-6}
Volts → DN	[DN/V]	$\mathbf{4.76 \times 10^3}$	$\mathbf{4.76 \times 10^3}$	$\mathbf{4.76 \times 10^3}$	$\mathbf{4.76 \times 10^3}$

Table 9: Analytic conversion at the central pixel compared with the saved outputs.

Quantity	VNIR-0	VNIR-1	VNIR-2	VNIR-3
Radiance L [mW/m^2/sr]	69.8533	200.3996	347.5041	408.5177
Electrons, analytic from L [e$^-$]	52676.1	205092.4	462601.4	594119.3
Electrons, `ism_toa_e` [e$^-$]	52673.4	205082.8	420000.0	420000.0
Electrons, analytic from optical TOA [e$^-$]	52673.4	205082.8	462581.2	594093.7
DN, analytic from L [DN]	758	2953	6661	8555
DN, `ism_toa` [DN]	804	2985	4095	4095
DN, analytic from detection TOA, saturated [DN]	804	2985	4095	4095
Rel. diff. electrons [%]	-0.005	-0.005	-9.209	-29.307
Rel. diff. DN [%]	6.069	1.084	-38.523	-52.133

The analytic values do not match the saved outputs exactly. Some steps of the ISM change the value of the signal but not its units.

- The MTF is applied between the irradiance and the photons. It explains the difference of $-0.005\,\%$ in the electrons of VNIR-0 and VNIR-1. If the chain starts from the optical TOA, which already includes the MTF, the electrons match `ism_toa_e`.
- The FWC limits the electrons to 420000 e$^-$. In VNIR-2 and VNIR-3 the analytic value is higher, so the saved value is the FWC.
- The PRNU, the dark signal and the bad pixels are applied after `ism_toa_e`. They change the DN by $+6\,\%$ in VNIR-0 and by $+1\,\%$ in VNIR-1. If the chain starts from the detection TOA, the DN match `ism_toa`.
- The digitisation rounds the signal and saturates it at $2^{12} - 1 = 4095$ DN. VNIR-2 and VNIR-3 are saturated at the central pixel.

The two most significant steps are the ones with the largest factors. They are the conversion from irradiance to photons, with a factor between 1.5×10^4 and 2.9×10^4, and the conversion from volts to DN, with a factor of 4.76×10^3. The first factor grows with the wavelength because a photon of longer wavelength carries less energy.

3.5 Theoretical question. Which MTF defines the system?

The system MTF is the product of the six contributors. The contributor with the lowest value at Nyquist drives the system MTF down the most. Table 7 shows two dominant terms.

- The detector MTF is $|\mathrm{sinc}(f_n)|$. It only depends on the pixel width. At Nyquist it is close to $2/\pi \approx 0.64$ in all the bands.
- The wavefront error MTF depends on $(W/\lambda)^2$. It is lower at short wavelengths. In VNIR-0 it is 0.58 and it is the lowest contributor. In the other bands it is between 0.67 and 0.76.

So the detector defines the system MTF in VNIR-1, VNIR-2 and VNIR-3, and the wavefront error aberrations define it in VNIR-0. The diffraction MTF stays above 0.93. The optics cut-off frequency $D/(\lambda f)$ is between 9 and 17 times the sampling frequency $1/w$. Nyquist is then at only 3 to 6 % of the cut-off frequency. Defocus, smearing and motion blur are close to one. Smearing only acts in the ALT direction. The system MTF at Nyquist is between 0.36 and 0.45. This value is high, so the system is prone to aliasing.

EODP-TS-ISM: PASS.

4 EODP-TS-L1B. Level-1B processing

4.1 Objective

The test validates the L1B module. The L1B converts the digital numbers of the ISM back into TOA radiances in two steps [3].

- Equalization. It removes the offset and the gain of each pixel: $DN_{eq} = (DN - eq_{add})/eq_{mult}$.
- Restoration. It applies the absolute radiometric gain of the band: $L_{L1B} = DN_{eq} \cdot g$.

4.2 Inputs and configuration

The input is the ISM output in `EODP_TER_2021/EODP-TS-L1B/input`. It contains `ism_toa` in DN and `ism_toa_isrf`, the radiance before the instrument effects. The equalization factors are the files `eq_add` and `eq_mult` of the auxiliary folder. They have one value per ACT pixel. The absolute gains are 0.0921, 0.0679, 0.0522 and 0.0478 mW/m^2/sr/DN for the four bands.

4.3 Execution

The script `harness/EODP-TS-L1B/test_l1b.py` runs the L1B twice. The first run has `do_equalization = False`. The second run has `do_equalization = True`. The second run is the one cross-validated.

4.4 Results

4.4.1 Cross-validation

Table 10 shows the comparison of the equalized DN and the L1B radiances with the reference. All the files are identical to the reference.

Table 10: L1B cross-validation with equalization. Maximum relative difference [%] per band.

File	VNIR-0	VNIR-1	VNIR-2	VNIR-3	Result
`l1b_toa_eq_`	0	0	0	0	Pass
`l1b_toa_`	0	0	0	0	Pass

4.4.2 Calibration plot

Figure 6 shows the central ALT line of three images. The black line is the TOA after the ISRF. It is the radiance that enters the instrument and it is taken as the truth. The red line is the L1B without equalization. The blue line is the L1B with equalization.

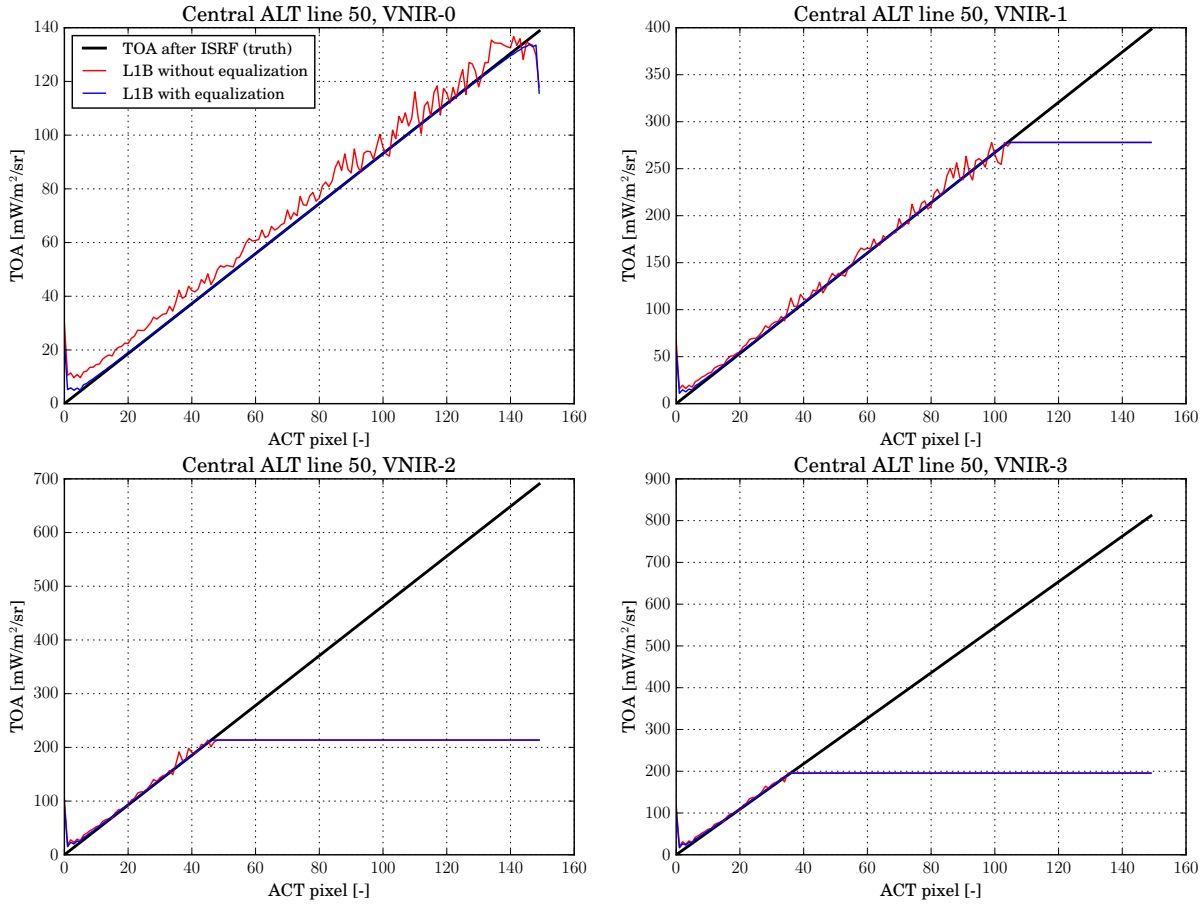


Figure 6: L1B. TOA after the ISRF, L1B without equalization and L1B with equalization. Central ALT line.

Without equalization the L1B shows the stripes of the PRNU and the DSNU. It is also shifted above the truth because the dark signal adds a positive offset. The shift is clear in VNIR-0, where the signal is low. With equalization the stripes and the offset disappear and the L1B follows the truth. Two differences remain. The first and last columns differ because of the MTF edge effect of the ISM. In VNIR-1, VNIR-2 and VNIR-3 the L1B becomes flat from a certain column. The L1B is "blocked" at a constant value. This happens because the ISM saturated these pixels at the FWC and at 4095 DN (Section 3). The information above saturation is lost and the L1B cannot recover it. The flat level is the radiance that corresponds to 4095 DN after the equalization and the gain.

Table 11 gives the median relative error with respect to the truth. Saturated pixels and the

two columns of each border are excluded. The median is used because the first columns have a radiance close to zero, where any small difference gives a large relative error. The equalization reduces the error in all the bands.

Table 11: L1B median relative error with respect to the TOA after the ISRF, and fraction of saturated pixels.

Band	Error without eq. [%]	Error with eq. [%]	Saturated pixels [%]
VNIR-0	7.494	0.233	0.0
VNIR-1	4.116	0.166	30.7
VNIR-2	4.216	0.946	68.7
VNIR-3	5.047	1.655	76.7

4.5 Theoretical question. What is calibration?

Calibration is the process that finds the parameters which relate the instrument output to the physical quantity observed. The instrument does not measure radiance. It produces digital numbers. Each pixel has its own response, with an offset from the dark signal and a gain from the PRNU. The calibration has two parts.

- Relative calibration. It finds the offset and the gain of each pixel, so that all the pixels give the same output for the same input. These are the equalization factors eq_{add} and eq_{mult}. The offset is measured with dark images, without illumination. The gain is measured with a uniform scene.
- Absolute calibration. It finds the gain g that converts the equalized DN into radiance. It is measured with a source of known radiance, such as an integrating sphere on ground or the Sun with a diffuser in flight.

The calibration parameters are measured in the calibration campaigns and stored in a characterisation database. The L1B processing does not compute them. It only applies them. Figure 6 shows the effect. Without the relative calibration the image has stripes and an offset. With it the L1B recovers the input radiance up to the saturation level.

5 EODP-TS-L1C. Level-1C processing

5.1 Objective

The test validates the L1C module. The L1C projects the L1B radiances onto the MGRS grid [3]. The steps are the following.

1. Build a 2D spline interpolant of the L1B radiance as a function of latitude and longitude (`bisplrep`).
2. Convert every L1B coordinate into an MGRS tile with a precision of 100 m and keep the unique tiles.
3. Convert each tile back to latitude and longitude. The `mgrs` library [1] returns the south-west corner of the tile.
4. Evaluate the interpolant at these coordinates (`bisplev`). The result is the L1C radiance.

5.2 Inputs and configuration

The inputs are the geolocation of the GM in `input/gm_alt100_act_150` and the L1B radiances in `input/l1b_output`. Both are matrices of 100 lines by 150 columns. The MGRS precision is 3, which gives tiles of 100 m.

5.3 Execution

The script `harness/EODP-TS-L1C/test_l1c.py` runs the L1C with both input folders separated by a comma. The L1C output is a list of points. Its order changes in each run because the tiles are stored in a Python set. For this reason each reference point is matched with the closest output point before the comparison.

5.4 Results

5.4.1 Cross-validation

Table 12 shows the comparison. The number of points is the same. Every reference point has an output point at zero distance, and the radiances are identical.

Table 12: L1C cross-validation after matching the points by coordinates.

Band	Points ref.	Points output	Max. distance [m]	Max. rel. diff. [%]	Result
VNIR-0	3129	3129	0	0	Pass
VNIR-1	3129	3129	0	0	Pass
VNIR-2	3129	3129	0	0	Pass
VNIR-3	3129	3129	0	0	Pass

5.4.2 L1B and L1C grids

Figure 7 shows both grids in the same plot. The L1B grid follows the geometry of the sensor. It is tilted with respect to the meridians because of the orbit inclination. The L1C grid is regular in UTM coordinates. Its points are aligned with the north and the east. The L1C grid is coarser. There are 3129 L1C points against 15000 L1B points.

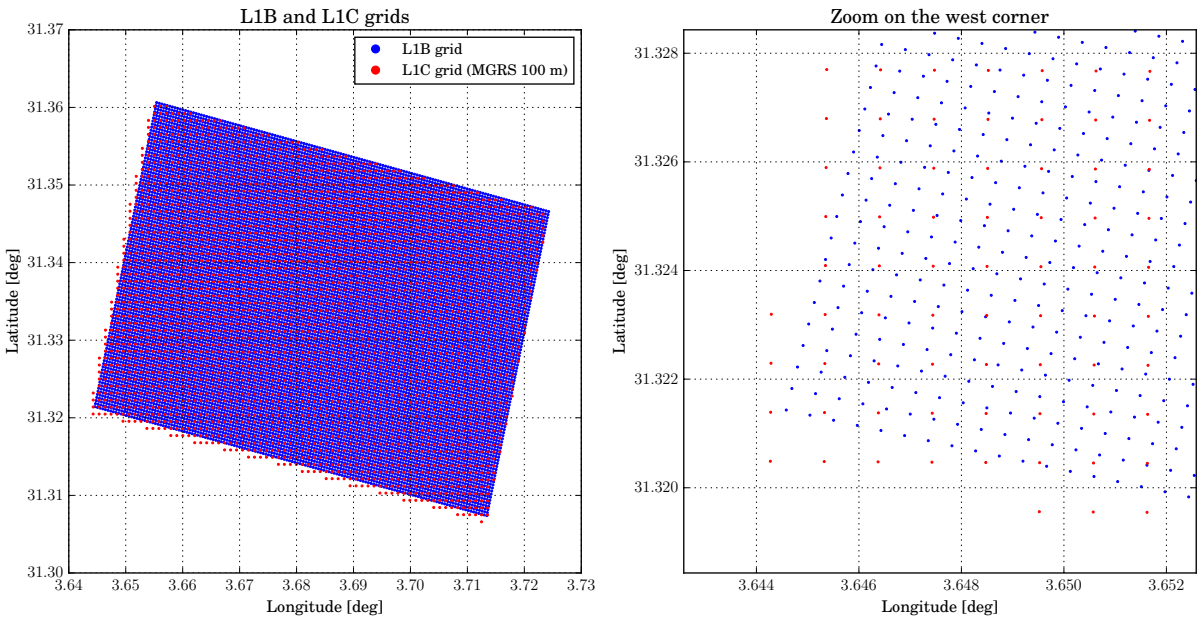


Figure 7: L1C. L1B and L1C grids. Full scene on the left and zoom on the west corner on the right.

5.4.3 Ground sampling distance

The GSD of the L1B is the distance on ground between consecutive pixels. It is computed with the Haversine formula along the central line (ACT) and the central column (ALT). Table 13 and Figure 8 show the result.

Table 13: L1B ground sampling distance.

Direction	Mean [m]	Min. [m]	Max. [m]
ACT	45.16	45.16	45.16
ALT	45.29	45.28	45.29

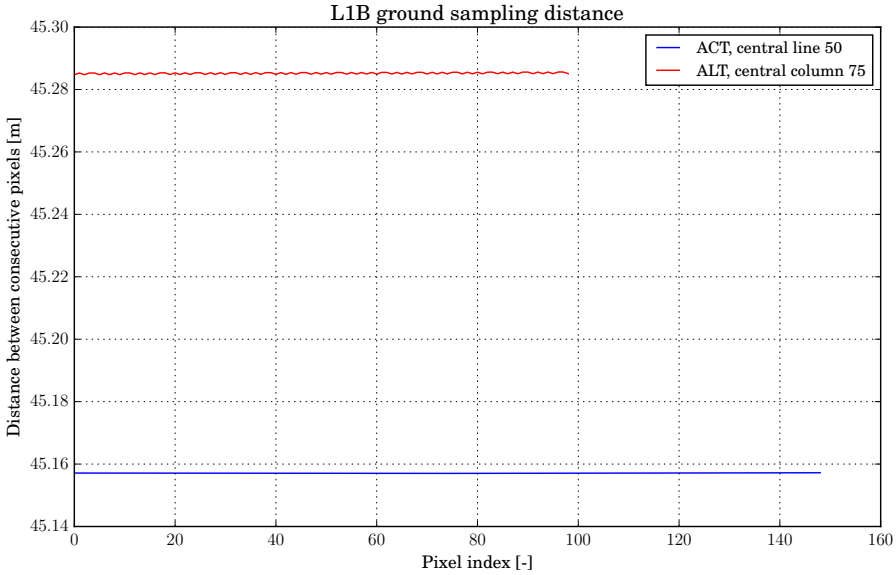


Figure 8: L1C. L1B ground sampling distance ACT and ALT.

The GSD is about 45 m in both directions and it is almost constant over the scene. Each L1B pixel covers about $45 \times 45 \approx 2045$ m^2. Each L1C tile covers $100 \times 100 = 10^4$ m^2. The ratio is 4.9. Then $15000/4.9 \approx 3070$ tiles cover the scene. The rest of the 3129 tiles are on the borders, where tiles are only partly covered.

5.4.4 L1C radiances

Figures 9 to 12 compare the L1B and L1C radiances on the map. The L1C keeps the gradient of the scene. On the west border some L1C values differ from their neighbours. The south-west corner of these tiles is outside the L1B footprint, so the spline extrapolates. In VNIR-2 two points have a negative radiance for this reason. A margin around the image or a check of the footprint would avoid it.

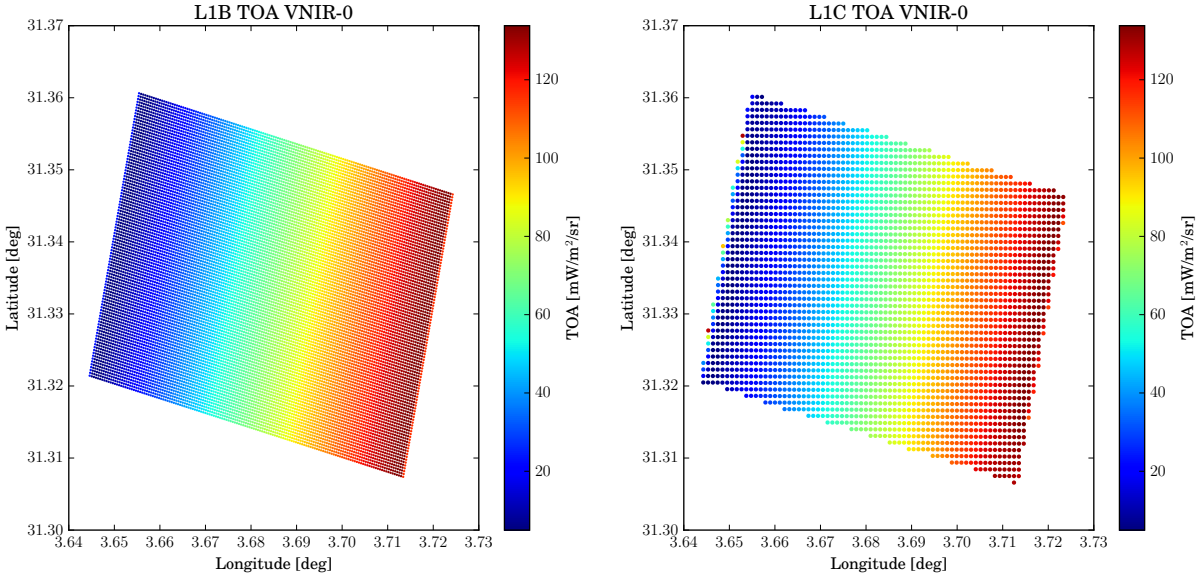


Figure 9: L1C. L1B and L1C TOA, VNIR-0.

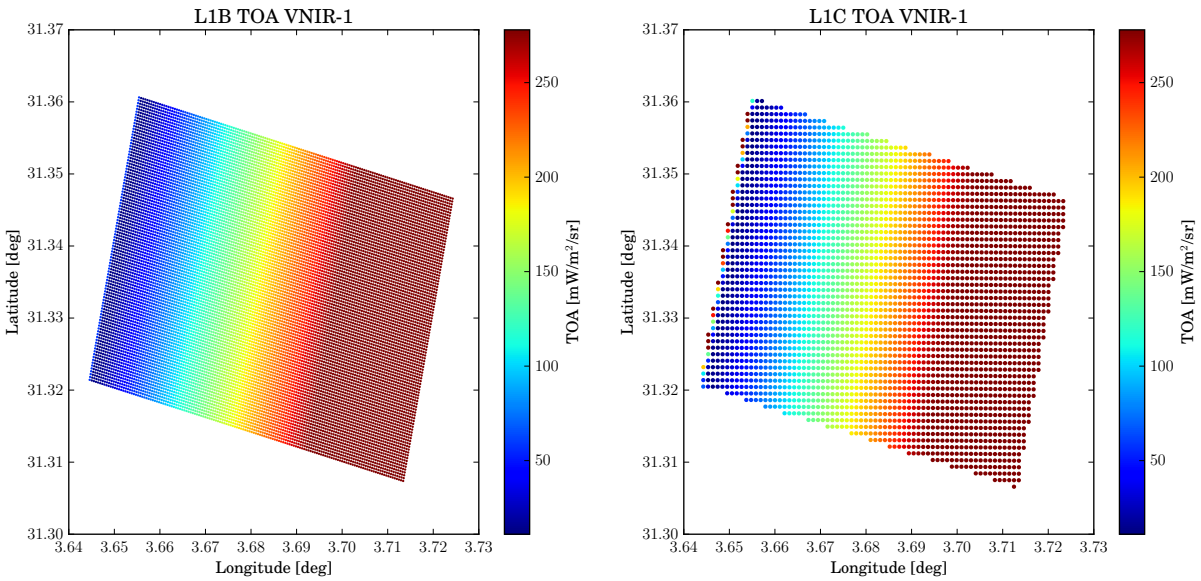


Figure 10: L1C. L1B and L1C TOA, VNIR-1.

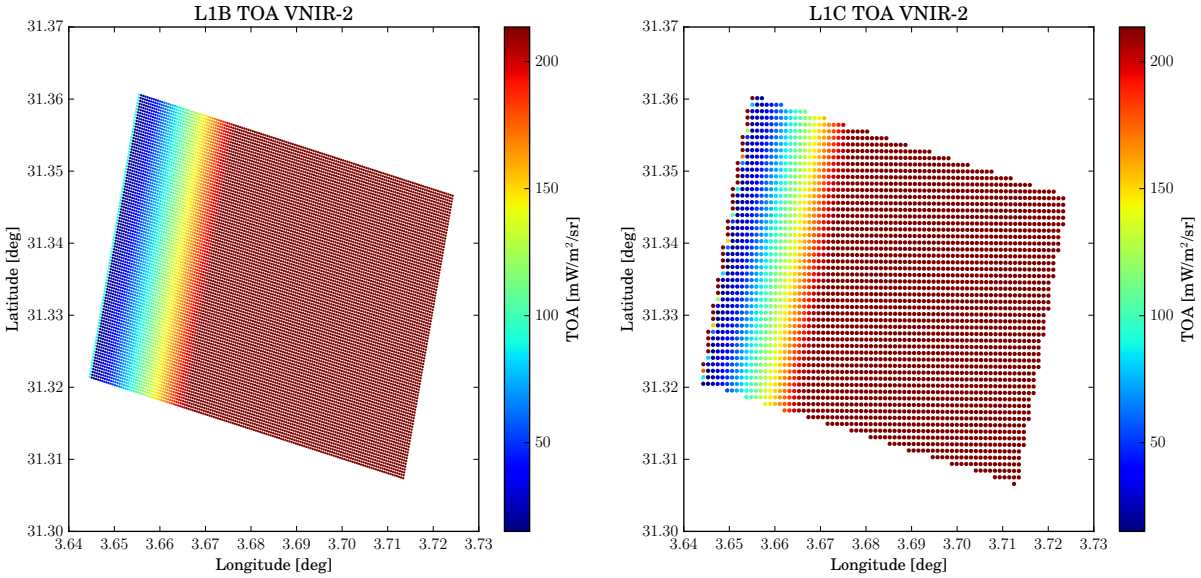


Figure 11: L1C. L1B and L1C TOA, VNIR-2.

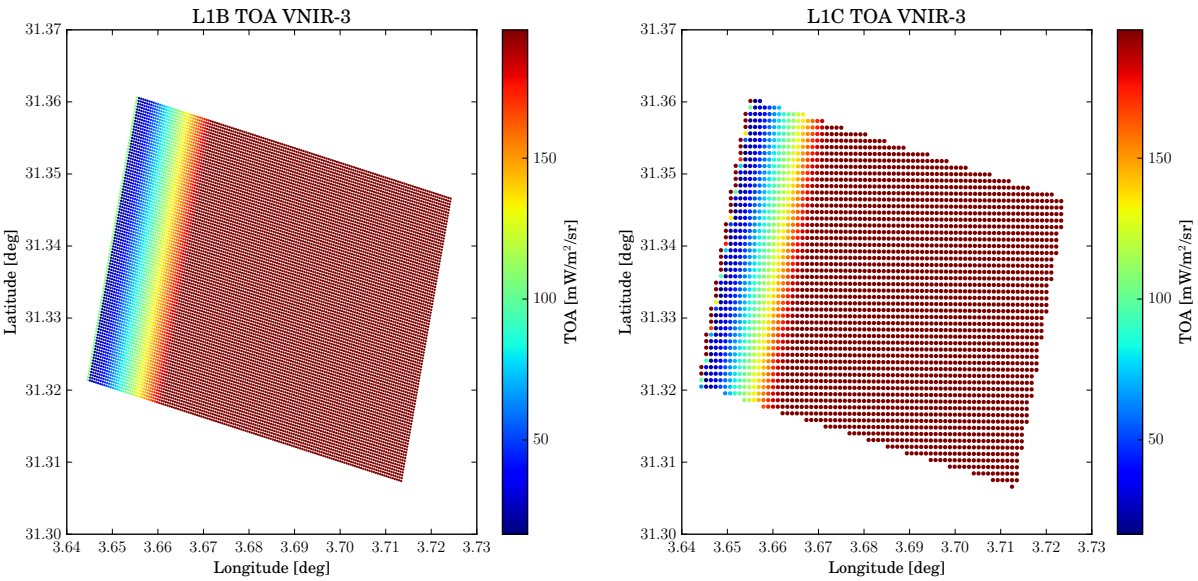


Figure 12: L1C. L1B and L1C TOA, VNIR-3.

5.5 Theoretical question. Processing levels and the need for the L1C

The data of an EO mission are processed in levels.

- Level 0. Raw instrument packets in time order, as received on ground.
- Level 1A. The packets are decoded into images in DN, with the time and the auxiliary data attached.
- Level 1B. The DN are converted into radiances with the calibration. The image keeps the geometry of the sensor and each pixel has its geolocation.
- Level 1C. The L1B radiances are resampled onto a fixed cartographic grid.
- Level 2. Geophysical variables retrieved from the L1, such as reflectance, vegetation indices or fluorescence.

The L1B grid is different for each acquisition. It depends on the orbit, the attitude and the terrain. Two L1B products of the same area cannot be compared pixel by pixel. The L1C solves this. All the products are projected onto the same grid, so they can be stacked in time, combined with other sensors and used to build maps. The L1B remains the most accurate radiance because it is not interpolated.

The grid used is the Military Grid Reference System (MGRS). It is based on the UTM projection, which divides the Earth into 60 zones of 6° in longitude. Each zone is a transverse Mercator projection with coordinates in metres. MGRS divides each UTM zone into latitude bands of 8° and squares of 100 km with two letters. Inside each square the easting and northing give the position. The number of digits fixes the precision. In the code `31REQ660644` is zone 31, band R, square EQ, easting 660 and northing 644 in units of 100 m.

The L1C has fewer points than the L1B because the MGRS tile of 100 m is larger than the GSD of 45 m. Several L1B pixels fall in the same tile and the tile is counted once. The ideal grid size would be the SSD of the sensor. The configuration uses 100 m, as the ATBD recommends [3].

6 EODP-TS-E2E. End-to-end test

6.1 Objective

The test runs the full chain on a real scene. The modules run in a row and each one reads the outputs of the previous one.

$$\text{SGM scene} \rightarrow \text{ISM} \rightarrow \text{L1B} \rightarrow \text{L1C}$$

The scene is a FLEX acquisition over the Pyrenees. The test checks that the chain runs without errors and that the L1B recovers the input radiance.

6.2 Inputs and configuration

The ISM reads the scene in `EODP-TS-E2E/sgm_out`. It has 951 lines, 525 columns and 650 wavelengths. The L1C reads the geolocation in `EODP-TS-E2E/gm_out` and the L1B output. The configuration is the one of `EODP-TS-E2E/config`. Table 14 shows the parameters that change with respect to the unitary tests. The radiometric errors are disabled, so the L1B does not need the equalization.

Table 14: E2E configuration. Parameters that differ from the unitary tests.

Parameter	Unitary tests	E2E
Pixel size w	30 µm	42 µm
Integration time t_{int}	6.72 ms	42.8 ms
Pupil diameter D	0.150 m	0.07565 m
Focal length f	0.5262 m	0.2345 m
PRNU, dark signal, bad and dead pixels	applied	not applied
L1B equalization	yes	no
L1B gain VNIR-0 [mW/m^2/sr/DN]	0.0921	0.00576
L1B gain VNIR-1 [mW/m^2/sr/DN]	0.0679	0.00425
L1B gain VNIR-2 [mW/m^2/sr/DN]	0.0522	0.00326
L1B gain VNIR-3 [mW/m^2/sr/DN]	0.0478	0.00299

6.3 Execution

The script `harness/EODP-TS-E2E/test_e2e.py` sets `EODP_CONFIG_DIR` to the E2E configuration and runs the three modules. The ISM took 195 s, the L1B 11 s and the L1C 61 s. The three modules finished without errors. As agreed in class, the report shows the L1B results only.

6.4 Results

6.4.1 Altitude

Figure 13 shows the variable `altitude` of `gm_out/geolocation.nc` on the map. The geometry is the same for the four bands, so there is a single plot. The altitude goes from 131 m to 2972 m. The Pyrenees cross the scene from west to east at about 42.6°N.

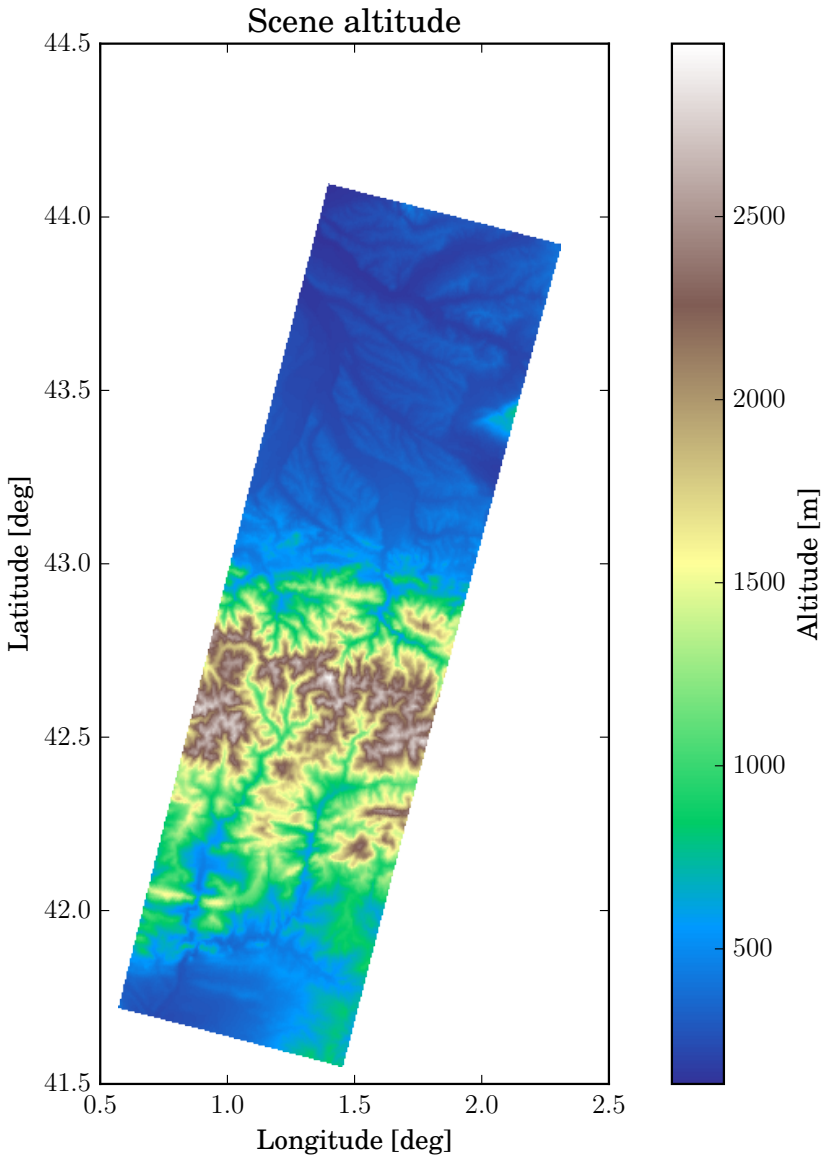


Figure 13: E2E. Altitude of the scene.

6.4.2 L1B radiances

Figures 14 to 17 show the L1B TOA of each band on the map, with its histogram. The histograms use bins of 0.1 mW/m^2/sr. The same structures appear in the four bands. The lowest radiances are in the mountains, on the slopes in shadow. VNIR-0 and VNIR-1 have the highest radiances, up to 13.7 mW/m^2/sr. VNIR-2 and VNIR-3 reach 7.1 and 8.5 mW/m^2/sr. The histograms are consistent with the maps. Most pixels are bright and a long tail on the left comes from the shadowed mountain areas.

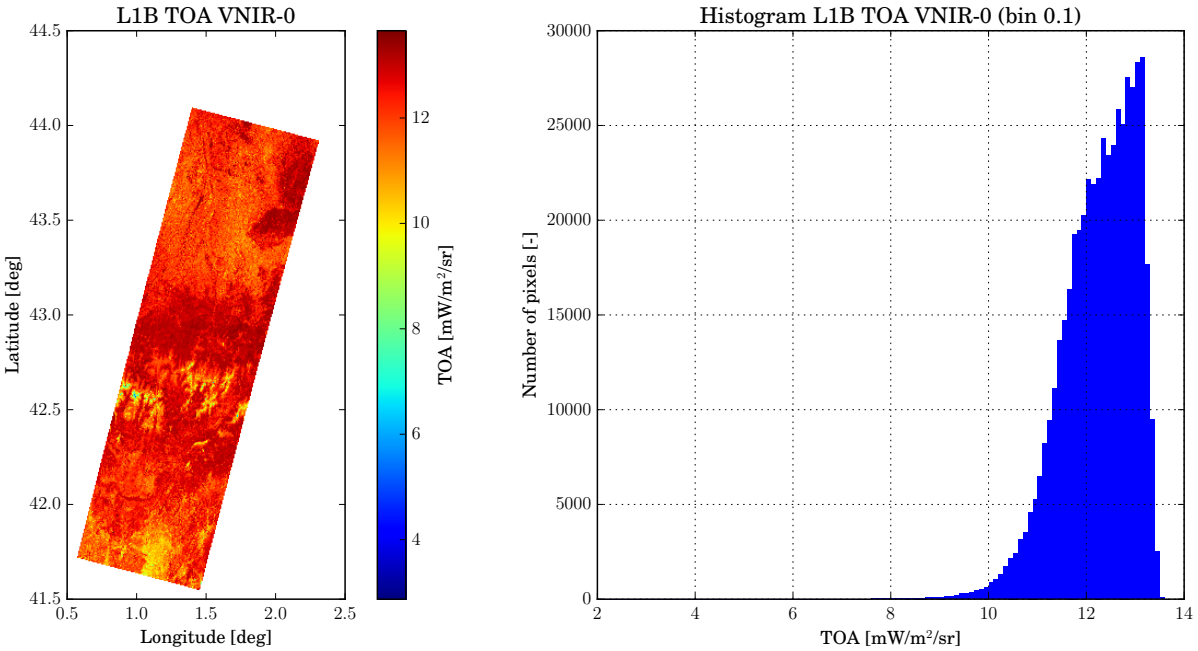


Figure 14: E2E. L1B TOA and histogram, VNIR-0.

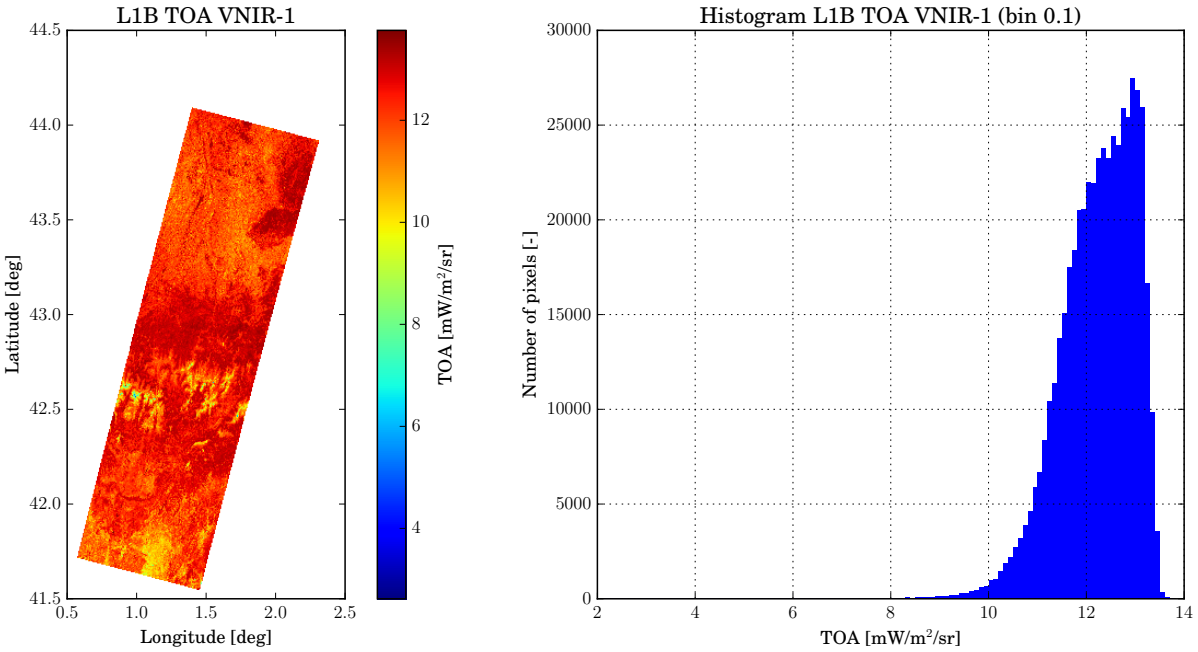


Figure 15: E2E. L1B TOA and histogram, VNIR-1.

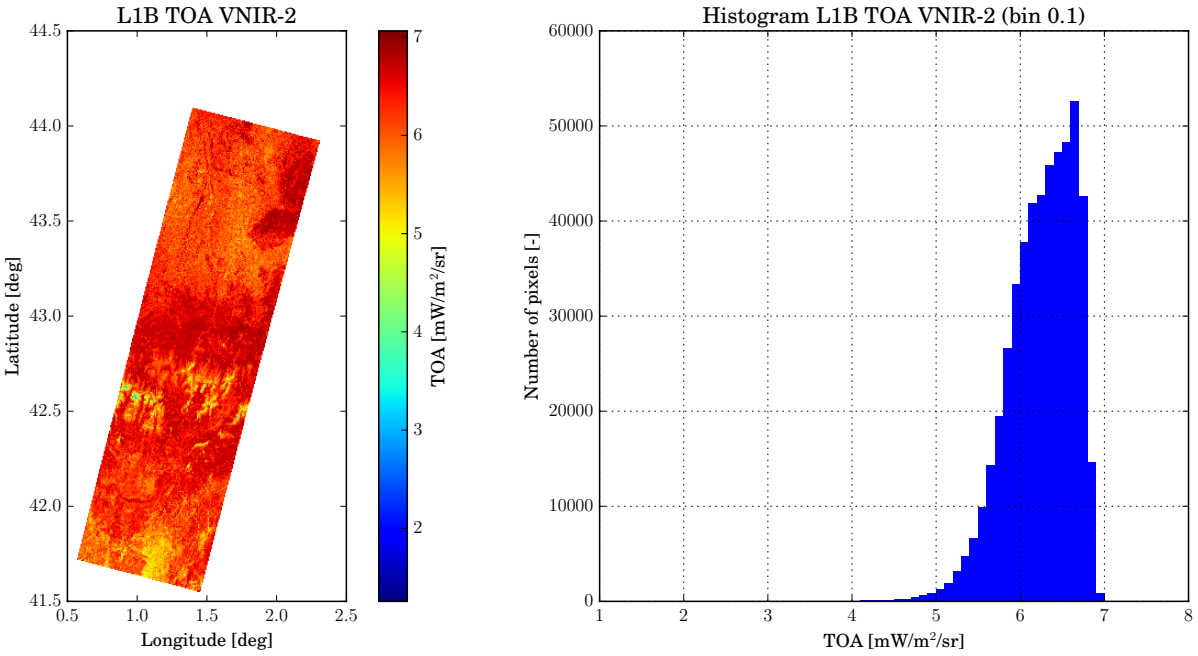


Figure 16: E2E. L1B TOA and histogram, VNIR-2.

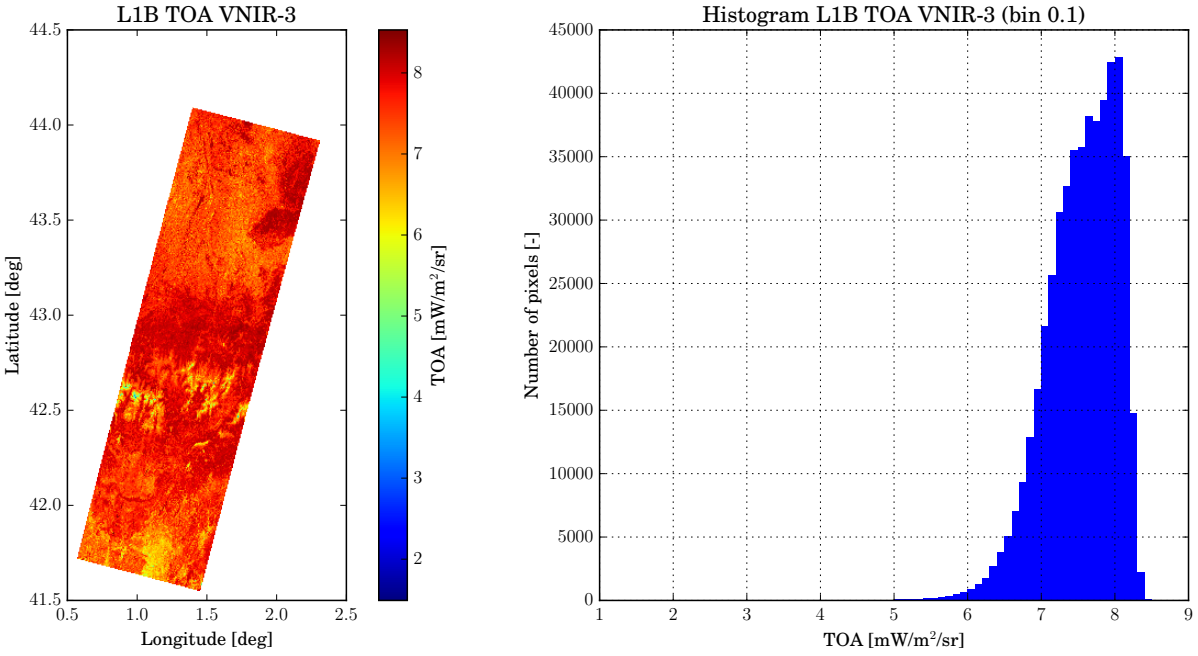


Figure 17: E2E. L1B TOA and histogram, VNIR-3.

6.4.3 Radiometric consistency

Without radiometric errors, the L1B should recover the radiance that enters the instrument. Table 15 compares the mean of `ism_toa_isrf` with the mean of `l1b_toa`. The difference is about $-0.02\,\%$ in all the bands. It comes from the MTF and the quantisation. No pixel is saturated.

Table 15: E2E. Mean radiance before the instrument and after the L1B [mW/m^2/sr].

Band	Mean ISRF TOA	Mean L1B TOA	Difference [%]	Saturated [%]
VNIR-0	12.2554	12.2529	-0.021	0.00
VNIR-1	12.2348	12.2326	-0.018	0.00
VNIR-2	6.2529	6.2520	-0.016	0.00
VNIR-3	7.5464	7.5453	-0.015	0.00

As an additional check, the ISM and L1B outputs were compared with the outputs provided with the E2E data. Table 16 shows that they are identical within 1.2×10^{-5} %.

Table 16: E2E. Comparison with the ISM and L1B outputs provided with the E2E data. Maximum relative difference [%] per band.

File	VNIR-0	VNIR-1	VNIR-2	VNIR-3	Result
`ism_toa_isrf_`	0	0	0	0	Pass
`ism_toa_optical_`	0	0	0	0	Pass
`ism_toa_e_`	0	0	0	0	Pass
`ism_toa_detection_`	0	0	0	0	Pass
`ism_toa_`	0	0	0	0	Pass
`l1b_toa_`	1.18×10^{-5}	1.19×10^{-5}	1.19×10^{-5}	1.19×10^{-5}	Pass

7 Conclusions

Table 17 summarises the results.

Table 17: Summary of the tests.

Test	Evidence	Result
EODP-TS-ISM	28 TOA files and 28 MTF files equal to the reference	Pass
EODP-TS-L1B	8 files equal to the reference	Pass
EODP-TS-L1C	4 files equal to the reference after matching the points	Pass
EODP-TS-E2E	Chain without errors. Mean radiance recovered within 0.02 %	Pass

The main findings are the following.

- All the outputs of the unitary tests match the reference with a relative difference below 10^{-3} %. Most of them are identical.
- The detector MTF defines the system MTF in VNIR-1, VNIR-2 and VNIR-3. The wavefront error aberrations define it in VNIR-0. The system MTF at Nyquist is between 0.36 and 0.45.
- The two largest unit conversion factors are the irradiance to photons and the volts to DN conversions. The MTF, the detection effects and the saturation explain why the analytic conversion differs from the saved outputs.
- In the unitary test scene VNIR-1, VNIR-2 and VNIR-3 saturate. The L1B cannot recover the radiance above the saturation level.
- The equalization removes the PRNU and DSNU stripes. The L1B then follows the input radiance.

- The L1B GSD is about 45 m. The L1C grid of 100 m has fewer points than the L1B. Near the borders the L1C spline extrapolates.
- The end-to-end chain works on a real scene. The L1B recovers the input radiance.

Two improvements are suggested. A margin around the image would remove the MTF edge effect of the ISM. A check of the L1B footprint in the L1C would avoid the extrapolated values on the borders.

References

[1] Howard Butler and contributors. mgrs. Python library for MGRS conversions. `https://pypi.org/project/mgrs/`. Version 1.5.0.

[2] Víctor Salas Gómez. EODP source code repository. `https://github.com/SalasGomez/EODP-submission`, 2026.

[3] Lucía Soto. EODP algorithm theoretical baseline document. Technical report, Universidad Carlos III de Madrid, 2022. Revision 1.4, 22/08/2022.

[4] Lucía Soto. Guide to the subject. earth observation data processing, 2026. Revision 3.0, 10/09/2026. Master in Space Engineering, UC3M.